

T-CNS §VIII preview: Numerical verification (Tier-1 rendition)

Hadi Hajieghrary

TABLE I
EXPERIMENTAL CONFIGURATION (TIER-1).

plant integrator	RK4 shapes + exact $SE(2)$ load step, $\Delta t = 0.01$ s (E10 s)
send epoch D_c	0.05 s; τ grid $\{0.05..1.6\}$ exact on the grid
fusion gain	$\alpha = \kappa D_c$, $\kappa = 2.0$ (record gain)
sensing	odom 50 Hz (surge noise $0.01+0.02 v $, sway 0.01, gyro 0)
maneuver	persistent turn $\eta(t) = 0.15 + 0.10 \sin(0.05t)$
formations	id-keyed draws, half-arc law matched to the multibody atta
metric	all-pairs mean $\ \text{Log}(G_i^{-1}G_j)\ ^2$, last 30% window
runs	E3b production: 1584 (12 formations $\times$ 10 seeds per $\tau + c$)

I. NUMERICAL VERIFICATION

All cells in this section run on the reduced plant inside protocol class $\mathcal{P}$ (uniform lags, zero jitter and loss, realized age asserted equal to τ per fusion event), with the estimator, metric functional, window rules, and fit code shared verbatim with the multibody companion — so agreement between the two papers is a statement about the algorithm, and disagreement localizes to the plant. Every claim below carries the pre-registered falsifier it was exposed to; three fired, and their firings are reported as findings. Every number traces to a seeded, versioned driver re-executed exactly by the released replay harness.

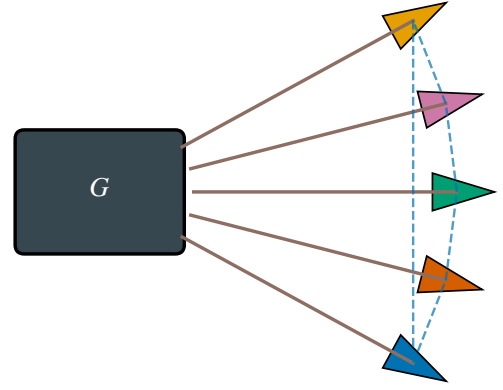
A. Plant, protocol, and information structure

Figure 1 defines the experimental object: N agents on taut unit cables to a common load, each carrying the shape pair $s_j = (\sigma_j, \sigma_{i,j})$, communicating over a graph whose algebraic connectivity λ_2 is a controlled variable (the C_5 cycle is the default; K_5 and the E6 sweep vary it). Figure 2 is the method pipeline actually executed: per agent, 50 Hz odometry drives a left-invariant propagation of the load-pose estimate through the slaved twist, a 20 Hz cable-direction channel updates the shape, and at every send epoch the sheaf-weighted fusion rule transports a τ -stale neighbor estimate through the shape-dependent edge map before combining. The information boundary is strict and machine-checked: each agent sees its own two sensors, its neighbors' stale estimates, and (in pinning cells only) one absolute beacon at one agent — the plant's truth state reaches the estimator only through the sensor models. Table I fixes every constant a reimplementation needs; Table II is the executed cell matrix of this section.

B. Gauge structure and pinning (E1; Thm. 5.1, Cor. 5.2)

On the triangle with generic shapes, the sheaf Laplacian is PSD with $\dim \ker L_{\mathcal{F}} = 3$ and the gauge section $(\text{Ad}_{m(s_j)}^{-1}c)_j$

Reduced plant: $N=5$ tow with C_5 communication



$s_j = (\sigma_j, \sigma_{i,j})$, cable l ; C_5 comms

Fig. 1. The reduced plant and its communication structure: shapes $s_j = (\sigma_j, \sigma_{i,j})$ on taut unit cables; the comms graph (dashed) is a controlled variable (the C_5 default shown; K_5 and the E6 sweep vary λ_2).

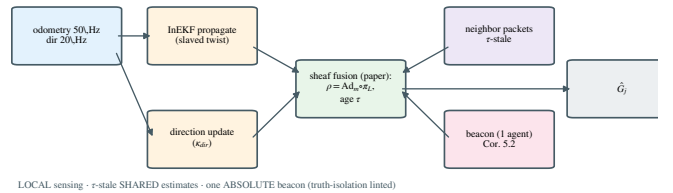


Fig. 2. The executed estimator pipeline and its information boundary: local sensing (odometry, cable direction), τ -stale shared estimates transported through $\rho = \text{Ad}_m \circ \pi_L$ with sheaf weights w_e , and at most one absolute anchor. Nothing else touches truth (lined).

annihilated to 1.3×10^{-15} ; σ_4/σ_3 confirms a clean spectral gap. Adding a single anchor potential collapses the kernel to zero and $\lambda_{\min}$ becomes the convergence rate — the pinning corollary as linear algebra (Fig. 3).

C. The theorem's object: amplitude, coefficient, switch-offs (E3a; Thm. 7.2)

The measured amplitude $\|\text{Log Hol}\|$ on the $m=2$ round trip fits slope 1.999 over $\tau \in [0.05, 1.6]$ s with coefficient

TABLE II
EXECUTED CELL MATRIX OF THIS SECTION.

cell	varies	falsifier(s)
E1	anchor on/off	kernel dim $\neq 3$
E3a + ext	τ , formation, switch-offs	C7a/C8/C13; C15
E3b	$\tau \times \text{arm} \times \text{formation} \times \text{seed}$	C7b ([CONJECTURAL])
E3c	ε off-symmetry, robust arm	C9b'/C9b/C9c
E2	κ , topology, $\ \delta\ $	C6, C19
E6	topology $\times N(\lambda_2)$	C10 (exploratory)
E10	$\Delta t (10\times)$	CI disjointness

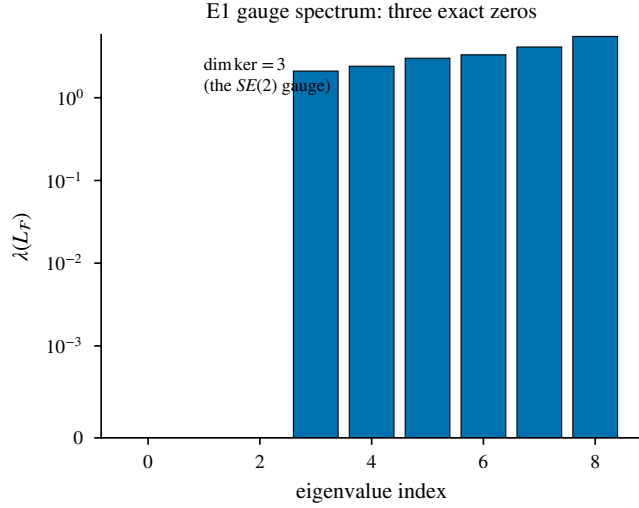


Fig. 3. Gauge spectrum: $\dim \ker L_F = 3$ without an anchor, collapsing to zero with one anchor whose $\lambda_{\min}$ becomes the rate. (The harmonic-section identity — kernel = body-framed images of one common load-frame perturbation — is the 1.3×10^{-15} annihilation check reported in the text.)

ratio $\|\text{Log Hol}\|/\tau^2\|[C_i, C_j]\| = 1.0000$ at the smallest τ ; the switch-off arms sit at machine zero ($s_i \equiv s_j$: $< 10^{-15}$; $\eta=0$: $< 10^{-16}$; $\xi=0$: exactly 0.0; explicit $\tau=0$ arm: exactly 0.0). Extension arms sharpen this: the slope is formation-invariant across 20 random taut formations (1.9993 ± 0.0000 , Fig. 7); the remainder constant is uniformly tight ($\sup_{s,\tau} \|R\|/\tau^3 \leq 0.0133$ over 220 draws); and a search for zero-commutator pairs at ≥ 1 rad shape separation found only pairs with $C_i = C_j$ to 10^{-16} (Fig. 8) — the protected class of Cor. 7.3 extends, empirically, to the level set of $s \mapsto C(s; \xi)$, where the full holonomy vanishes at all orders.

D. The conjectured closed-loop floor, falsified in its own regime (E3b)

With the full DIEKF- Σ on the reduced plant integrating the shape ODEs along the persistent-turn reference (1584 runs; 12 formations $\times$ 10 seeds per τ point; controls on matched draws), the disagreement excess over the frozen-shape control fits exponent 1.101 [1.076, 1.125] (formation-cluster CI [1.070, 1.132]); the [CONJECTURAL] extension of the floor to the noisy closed loop is falsified at these scales, coherently with the multibody companion (cross-tier exponent difference $+0.023 [-0.012, +0.057]$). The mechanism check passes — the straight-tow control sits $\approx 5\times$ below the paper

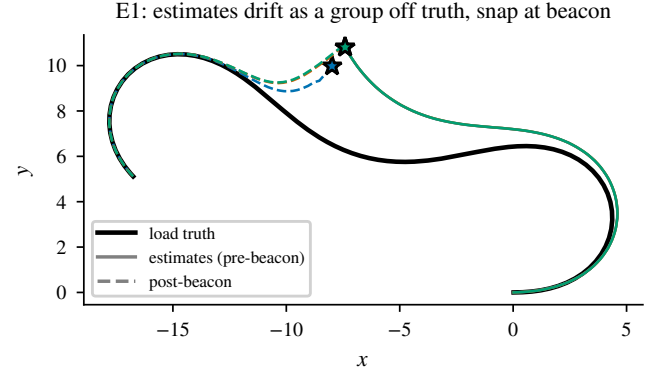


Fig. 4. The gauge phenomenon as trajectories: with no anchor the agents' load-pose estimates (colours) track each other but drift as a rigid group off the truth (black); at the beacon (stars) they snap back (dashed).

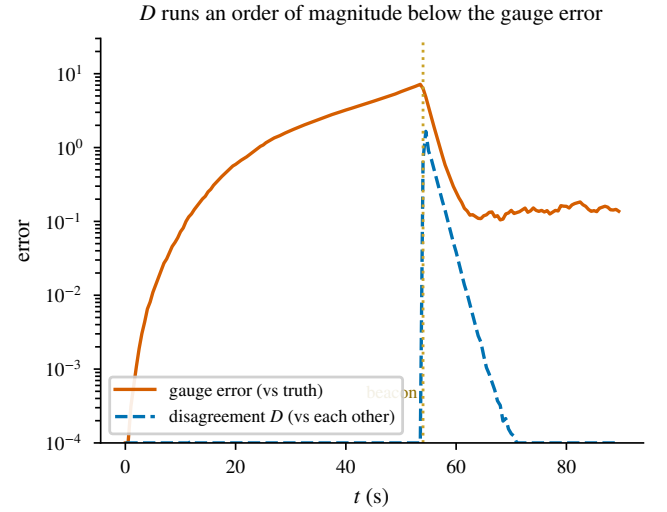


Fig. 5. The two error objects are distinct: the disagreement D (agents vs. each other) runs about an order of magnitude below the gauge error (agents vs. truth); one anchor collapses both (Thm. 5.1, Cor. 5.2).

arm and the frozen-shape control $\approx 15\times$ below, at every τ — and the executed null pins the harness: shapes frozen and noise off give $D_{ss} \sim 10^{-30}$.

Figures 10 and 11 expose the *distributions* the fitted exponent summarizes. At the reference staleness $\tau = 0.4$ (Fig. 10) the arm separation is distributional, not just in medians: the paper-rule and frozen-shape CDFs do not overlap at any quantile. The paper-rule spread across the τ grid (Fig. 11) is $\approx 1.6\times$ interquartile, $\approx 3.7\times$ between the 5th and 95th percentiles — comparable to one τ octave, which is exactly why the campaign pre-registered seed-paired estimators for the ε -response (below) and why unpaired CIs were declared under-powered.

E. Symmetry: two metrics, two exponents (E3c; Cor. 7.3)

The amplitude departure from the symmetric class is first-order in ε (measured 1.006; base commutator exactly zero). The closed-loop floor excess under the seed-paired estima-

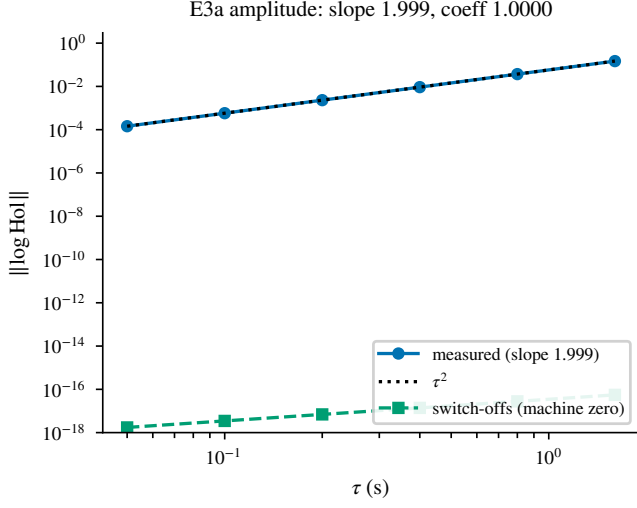


Fig. 6. F4a: the leading-order, deterministic holonomy amplitude of Thm. 7.2 — slope 1.999, coefficient 1.0000, switch-offs at machine zero.

E3a: 20 formations overlaid; slope 1.9993 for every one

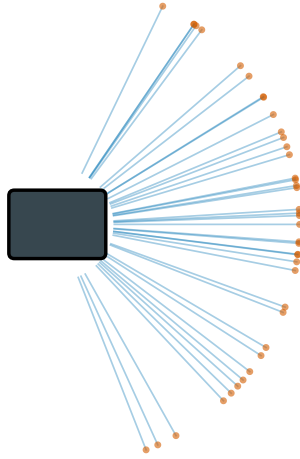


Fig. 7. E3a formation invariance: 20 random taut formations overlaid — the geometry varies widely, yet every one fits slope 1.9993. The τ^2 order is a structural constant.

tor fits ε -exponent 1.58 [1.44, 1.84] at every τ — excluding both the conjectured 2 and 1, with segment slopes rising $1.2 \rightarrow 2.1$ up the ladder: a linear-plus-quadratic mixture that the registered single-power model mis-specifies (reported per the outcome ladder). The robust $\geq 10\times$ suppression falsifier trips ($2.75 \times [1.94, 3.89]$ under 20% jitter and 10% loss): large symmetric protection is a property of the amplitude object only (Fig. 14). Figure 13 shows the tripping row at run level — the full generic and symmetric D_{ss} distributions under the robust arm. The registered statistic is the ratio of arm *means*, $2.75 \times [1.94, 3.89]$ (the boxes display medians, whose ratio is $3.2\times$):

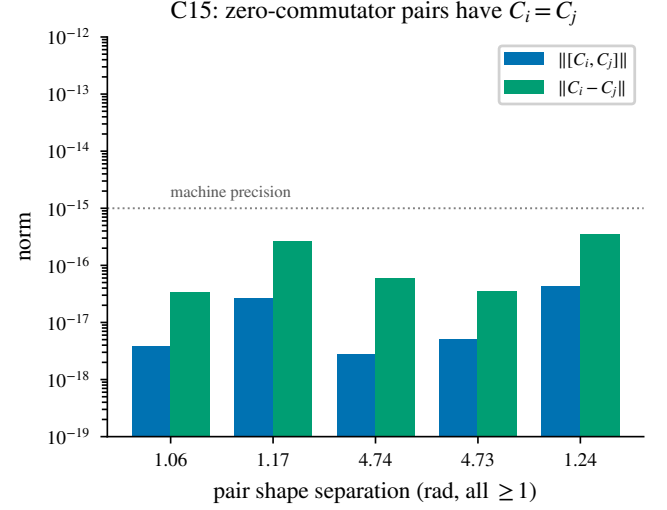


Fig. 8. E3a level-set protection (C15): the five zero-commutator pairs at ≥ 1 rad separation all satisfy $C_i = C_j$ to 10^{-16} — the protected class is a level set, not just coincident shapes.

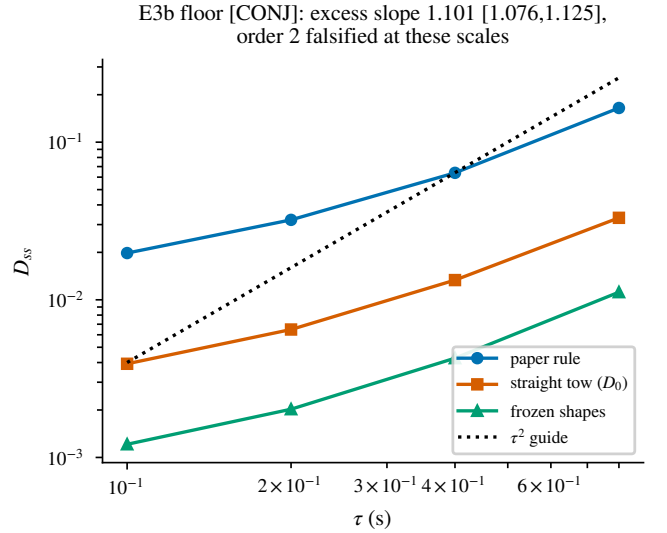


Fig. 9. Numerical evidence for the conjectured stochastic steady-state floor (IES §10(ii)); Theorem 7.2 is leading-order and deterministic and is tested in Fig. 6. Fitted excess exponent 1.101 [1.076, 1.125]: the conjectured order 2 is excluded (and so is 1); the frozen+noise-off regression reproduces the executed null ($D_{ss} \sim 10^{-30}$).

real, but an order short of the registered $10\times$, and explicitly outside protocol class $\mathcal{P}$ (heterogeneous realized ages), so it characterizes robustness rather than testing the theorem.

F. Contraction in its own domain (E2; Thm. 6.3)

Step-perturbing all agents and fitting the decay of the gauge complement: the rate scales with $\kappa\lambda_2$ at slope 1.403 across a $4\times$ gain range and two topologies, with $\mu = -0.062$ topology-independent. The log-linearity remainder vanishes *faster* than the registered second order (scaling-pair exponent 2.48 [2.32, 2.65]): the falsifier band is excluded on the favorable side

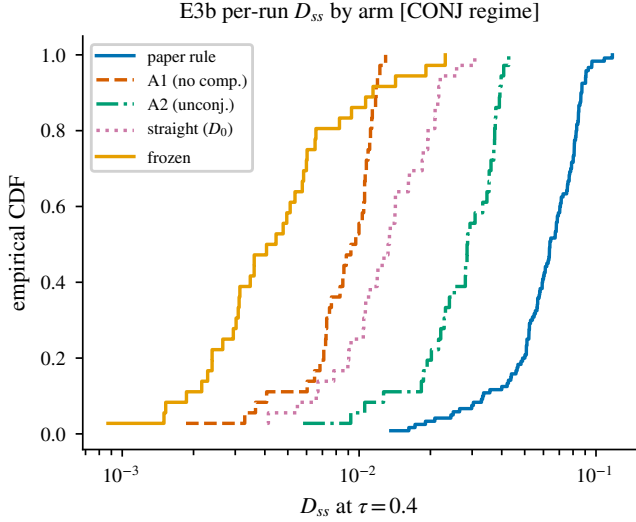


Fig. 10. Per-run D_{ss} CDFs by arm at $\tau = 0.4$ (1584-run production record). D_{ss} is the stochastic steady-state object [[CONJECTURAL] regime] — measured, not a test of Thm. 7.2.

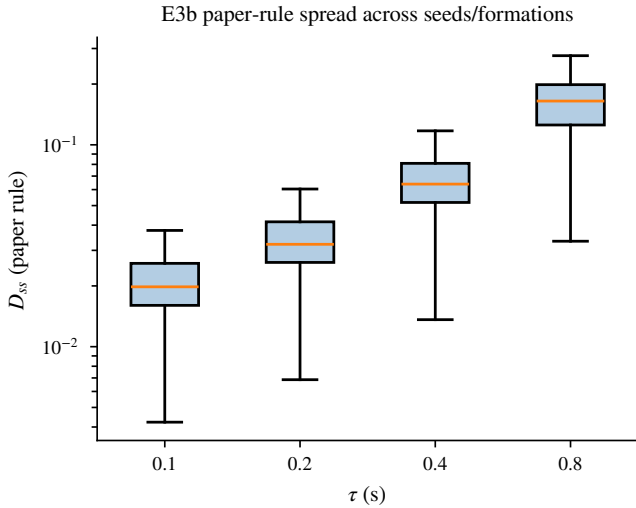


Fig. 11. Paper-rule D_{ss} spread across seeds and formations per τ ([[CONJECTURAL] regime]).

— reported as tripped, with the corrected order left to a restatement.

G. Numerics invariance (E10)

The excess exponent is invariant across integrator steps $\Delta t \in \{0.001, 0.0025, 0.005, 0.01\}$ s (all CIs share $[1.185, 1.256]$), with the coefficient converging mildly under refinement: no exponent above is an artifact of the discretization (Fig. 19).

H. Connectivity and the floor (E6; exploratory)

Pre-registered as exploratory (C10, no falsifier): sweeping topology and fleet size over cycle/path/star/complete at $N \in \{4, 6, 8\}$, connectivity suppresses the floor by up to $5\times$ at

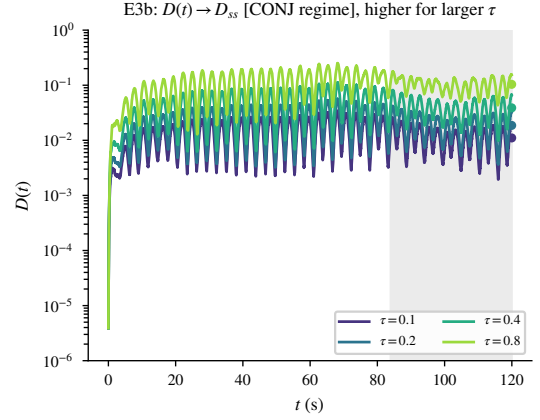


Fig. 12. The floor's dynamics: the disagreement $D(t)$ climbs from a perturbation to its steady state D_{ss} (dots) over a sweep of staleness τ , higher for larger τ , inside the shaded evaluation window. This is the measurement protocol for the [[CONJECTURAL]] steady-state object; it is not a test of Thm. 7.2, whose object is the amplitude of Fig. 6.

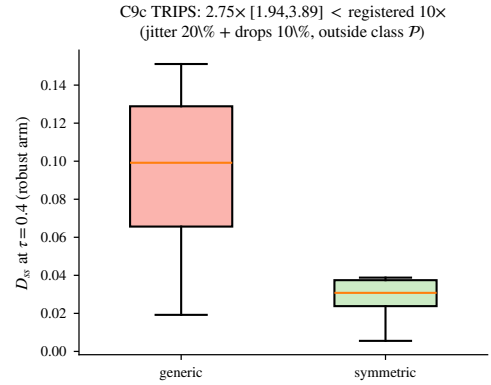


Fig. 13. The C9c trip at run level: generic vs. symmetric-class D_{ss} under 20% jitter + 10% drops (outside class $\mathcal{P}$). Median ratio $2.75 \times [1.94, 3.89]$ — protection persists but falls an order short of the registered $10\times$.

moderate staleness, and the benefit collapses ($\leq 2\times$) at high staleness, with complete graphs the most τ -sensitive (Fig. 20). Read forward, this is a design observation: buying links buys floor suppression only while the staleness is moderate — at high τ every topology pays the same curvature price. The anchoring-side counterpart (whether connectivity buys *pin-rate*) is a multibody question and is answered negatively in the companion.

I. Switch-off ladder and the transport-rule ablation

Figure 21 adds the closed-loop transport-rule ablation (E3b arms) as median D_{ss} per arm and staleness, and it contains a finding we flag rather than smooth over: on *this* plant, the A1 ablation (no lag compensation — the rule-registry semantics; A2 is the un-conjugated transport) sits *below* the paper rule on D_{ss} — the ordering inverts relative to the multibody plant, where the transport ablations degrade correctness $1.7\text{--}8.2\times$. The resolution is the metric, not the rule: D_{ss} is a *disagreement* functional, and agreement is purchasable by groupthink; it never ranks rules. Rule ranking belongs to correctness metrics

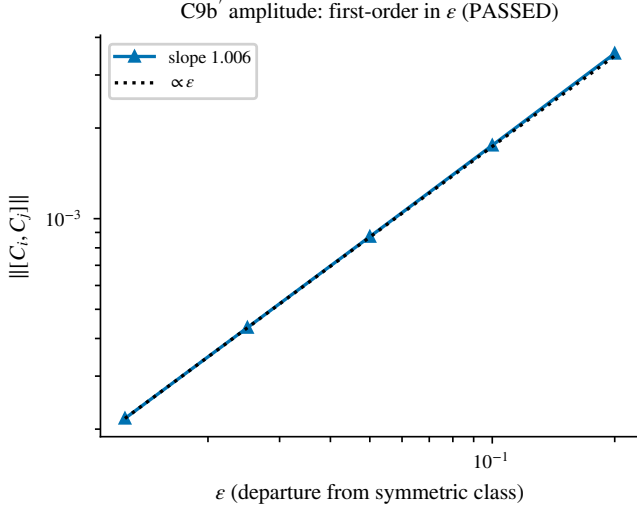


Fig. 14. Symmetry, the amplitude object (C9b'): the departure from the symmetric class is first-order in ε (slope 1.006), the base commutator exactly zero — PASSED.

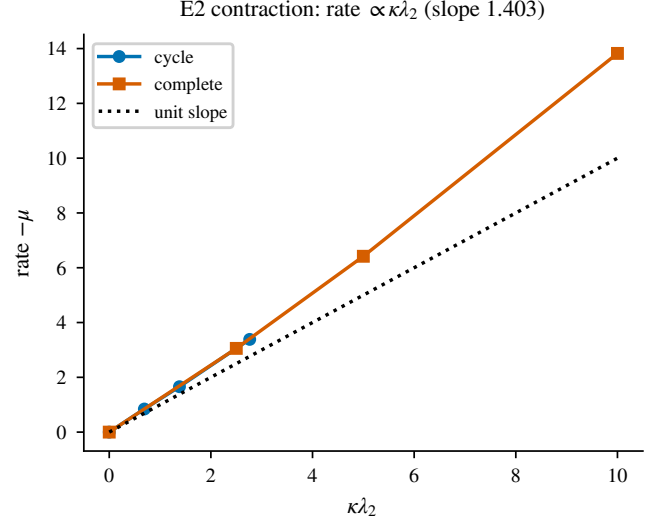


Fig. 16. F8: rate versus $\kappa\lambda_2$ across two topologies, slope 1.40 (falsifier < 0.5); μ calibrated at $\kappa = 0$, so the identity line is anchored, not predicted; the nonlinear extension is open.

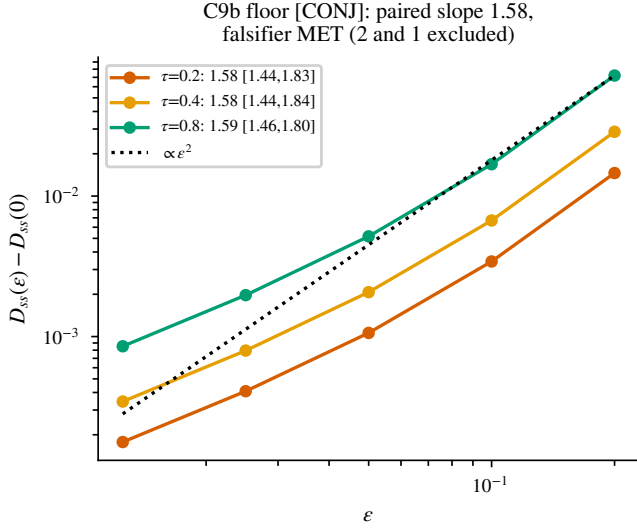


Fig. 15. Symmetry, the closed-loop floor (C9b, [[CONJECTURAL]] regime): the seed-paired ε -exponent is 1.58 at every τ , excluding both 2 and 1 — falsifier MET, a linear-plus-quadratic mixture the registered single-power model mis-specifies. Never mixed with the amplitude object.

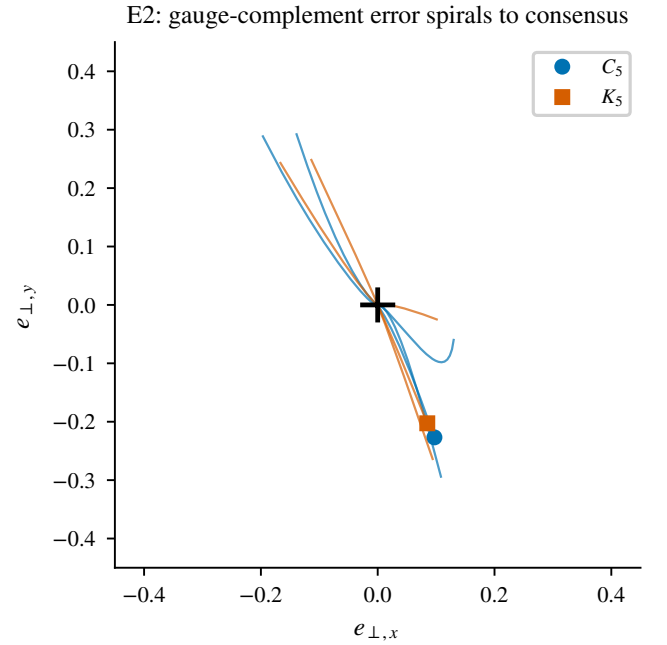


Fig. 17. Contraction as trajectories: after a step perturbation, each agent's gauge-complement error spirals into consensus (the origin), faster under the more connected K_5 than the C_5 cycle (LTV/frozen regime).

against truth, which exist only on the companion's plant, where the ordering is clean. We show the inversion because suppressing it would misrepresent what D_{ss} measures.

J. The adjudicated ledger, in one figure

Figure 22 renders the section's adjudicated falsifier rows — every quantitative falsifier this section reports — with registered references and verdicts: the statistical integrity statement. (The structural E1 falsifier, kernel dimension $\neq 3$, is pass/fail rather than an estimate and is reported in Sec. E1.) The proved chain's rows (C7a slope, C9b', C6, E10) are green; the [CONJECTURAL] regime's rows are red where the falsifier fired (C7b, C9b, C9c), exactly as reported in the

subsections above; C19 is orange: tripped in the favorable direction. Nothing is restated as a pass, and the one-axis view makes the epistemic split of the paper visible at a glance: the theorem's objects survive every registered attack; the conjectured stochastic extension does not, and is reported as measured.

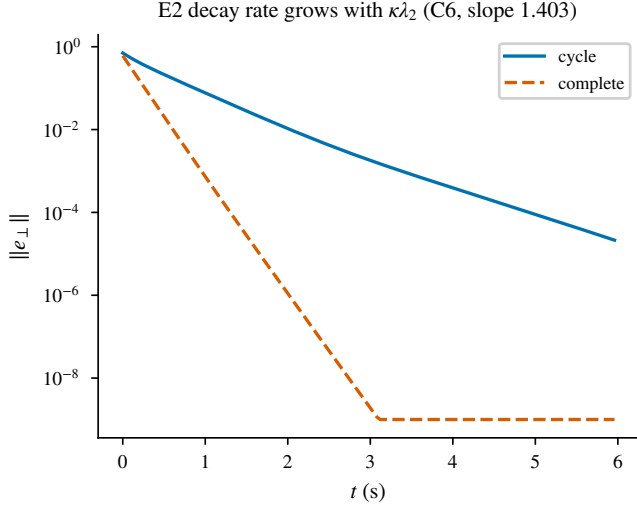


Fig. 18. The gauge-complement decay: its rate scales with $\kappa\lambda_2$ (C6, slope 1.403), faster on the more connected graph.

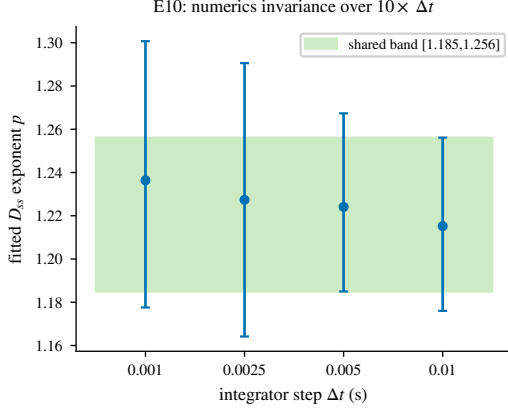


Fig. 19. E10: the fitted exponent across a $10\times$ range of integrator steps; all CIs share the green band $[1.185, 1.256]$.

K. Supplementary movies

Three supplementary movies accompany this section; each is generated by a seeded, versioned script from the same estimator code as the campaign. *M1* (*gauge_drift.mp4*) — the E1 phenomenon: three agents fuse without an anchor, remain mutually consistent, and drift *as a group* along the $SE(2)$ gauge; a beacon at one agent collapses the gauge and all three snap to truth. The two curves it traces — disagreement vs. truth error — are the two different objects this paper refuses to conflate. (Playback: 8 fps, time-compressed; the on-screen clock is authoritative.) *M2* (*contraction.mp4*) — the E2 mechanism: identical step perturbations decaying under C_5 and K_5 side by side; per-agent gauge-complement errors shrink toward consensus at rates ordered by $\kappa\lambda_2$ (C6’s tested prediction; LTV/frozen hedge carried on-screen). (Playback: 15 fps; 6 s of simulated decay over ≈ 8 s of video.) *M3* (*floor_protocol.mp4*) — the E3b *protocol*: the reduced plant executes the persistent turn while $D(t)$ for two stalenesses rises to its steady state inside the declared evaluation window. (Playback: 25 fps; 120 s

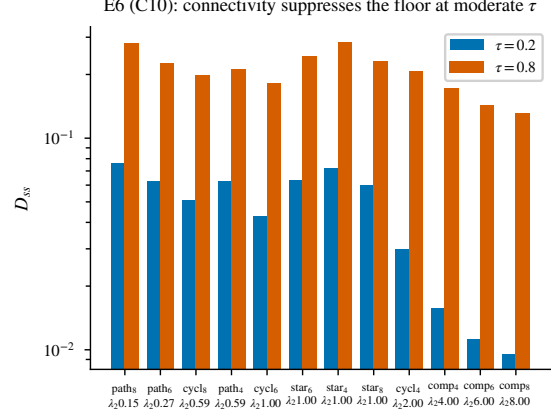


Fig. 20. E6 (exploratory, C10): floor vs. topology $\times N$ ordered by λ_2 , at moderate ($\tau=0.2$) and high ($\tau=0.8$) staleness. Connectivity suppresses the floor $5\times$ at moderate τ ; the benefit collapses at high τ .

TABLE III
SWITCH-OFF LADDER (TIER-1 ROWS). EACH ARM KILLS THE EFFECT ON COMMAND OR SHIFTS ITS ORDER AS PREDICTED.

arm	prediction	measured
generic slope	2	1.999 (formation-invariant 1.9993 ± 0.0000)
coefficient ratio ($m=2$)	1	1.0000
$s_i \equiv s_j$	0	$< 10^{-15}$
level set $C_i = C_j$, $\text{sep} \geq 1$	—	full holonomy $\equiv 0$ (observed)
$\eta = 0$	0	$< 10^{-16}$
$\xi = 0$	0	0.0
$\tau = 0$ (explicit arm)	0	0.0
frozen shapes, closed loop	null	$D_{ss} \sim 10^{-30}$

of simulation in ≈ 24 s, $\approx 5\times$ compression.) The on-screen caption carries the registry hedge verbatim: D_{ss} is measured in the [[CONJECTURAL]] regime; the movie demonstrates the protocol and is not a test of Thm. 7.2.

L. Reproducibility and artifact registry

Every number in this section traces to a seeded, versioned driver and a committed record; the released `campaign_replay.py` re-executes per-run probes that must match the records exactly (Tier-1 to 10^{-9}). Table IV is the executed registry: cell, driver, committed record, and probe class. The figure generator (`tier1_sheaf/campaign/paper_artifacts.py`) re-derives the extension datasets seed-exactly at import and refuses to plot on mismatch, so the figures cannot drift from the records. The repository is released with the paper (archived tagged release with DOI at submission); reproduction commands are in its root README. The multibody campaign, its falsifications, and the cross-tier equivalence tests appear in the RA-L companion.

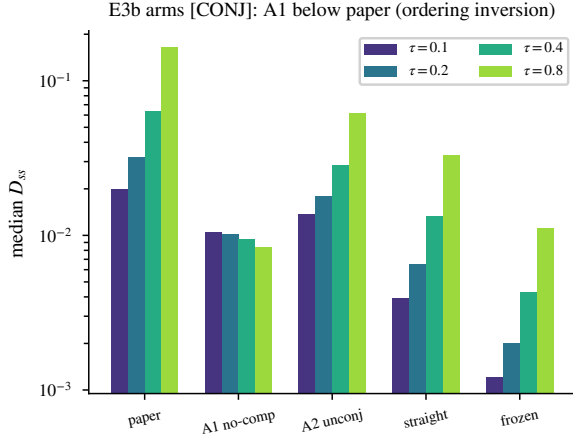


Fig. 21. E3b transport-rule and control arms, median D_{ss} by staleness [[CONJECTURAL] object]. The Tier-1 A1-below-paper inversion is shown, not smoothed: D_{ss} is a disagreement metric and never ranks rules — ranking is done on the companion’s truth-referenced metrics. The noise-off frozen null ($D_{ss} \sim 10^{-30}$) lies 27 orders below the axis and is reported in the table.

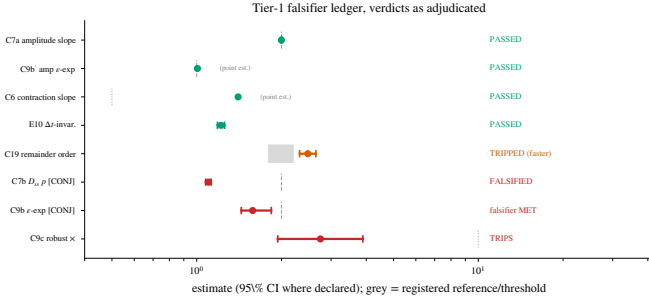


Fig. 22. The complete Tier-1 falsifier ledger, verdicts as adjudicated: green passed, red falsified/met ([CONJECTURAL] rows — measured slopes reported as measured), orange tripped favorably. Grey marks the registered references; the evidence record is named per row.

TABLE IV
EXPERIMENT REGISTRY (CELL → DRIVER → RECORD → PROBE).

cell	driver (tier1_sheaf/)	record	probe
E1	experiments/e1_gauge.py	e1_gauge.csv	exact
E3a	experiments/e3a_amplitude.py	e3a_amplitude.csv	exact
E3a-ext	campaign/e3a_extension.py	e3a_extension.json	exact
E3b	campaign/e3b_production.py	e3b_production.json (1584)	exact
E3c	experiments/e3c_symmetry.py + campaign/e3c_c9b_v2.py	e3c_*.json	exact
E2	experiments/e2_contraction.py	e2_contraction.json	stat
E6	campaign/e6_topology.py	e6_topology.json	exact
E10	campaign/e10_dt_sweep.py	e10_dt_sweep.json	exact
movies	campaign/*_movie.py, experiments/anim_gauge.py	*.mp4	seeded